

Software

The application program is developed in MATLAB R2014a with the Simulink toolbox. In order to build a Simulink model, 'Arduino IO Library' is needed, which can be obtained by installing 'Legacy MATLAB and Simulink Support for Arduino package' as explained below:

- Download Legacy MATLAB and Simulink Support for Arduino package from the MathWorks website. Extract ArduinoIO compressed folder
- From this folder, copy the pde folder and paste it in *C:\Program Files (or Program files x86)\Arduino\Examples* (path may be different depending on the installation directory of the IDE)
- After pasting the pde folder in the correct location, open Arduino IDE. If you have pasted the folder in the correct location, you would find pde in File→Examples→pde
- Open adios code from File→Examples→pde
- Connect the Arduino Uno board to your PC. From the device manager, note the COM port at which the Arduino Uno board is installed
- From Tools menu in Arduino IDE, select the board as *Arduino Uno* and select the COM Port number noted earlier
- Upload adios code to the Arduino Uno board by pressing the upload button in IDE
- Copy the entire contents of the extracted ArduinoIO folder to a folder in *My Documents* (for Windows PC). This step may be different if you are using the latest version of MATLAB
- Open Simulink LM35.slx source code file from MATLAB. The screenshot of the designed Simulink model is shown in Figure 3
- Double click on the block 'Arduino IO Setup' and change the serial (COM) port number (set as 'COM7' here) with the corresponding port

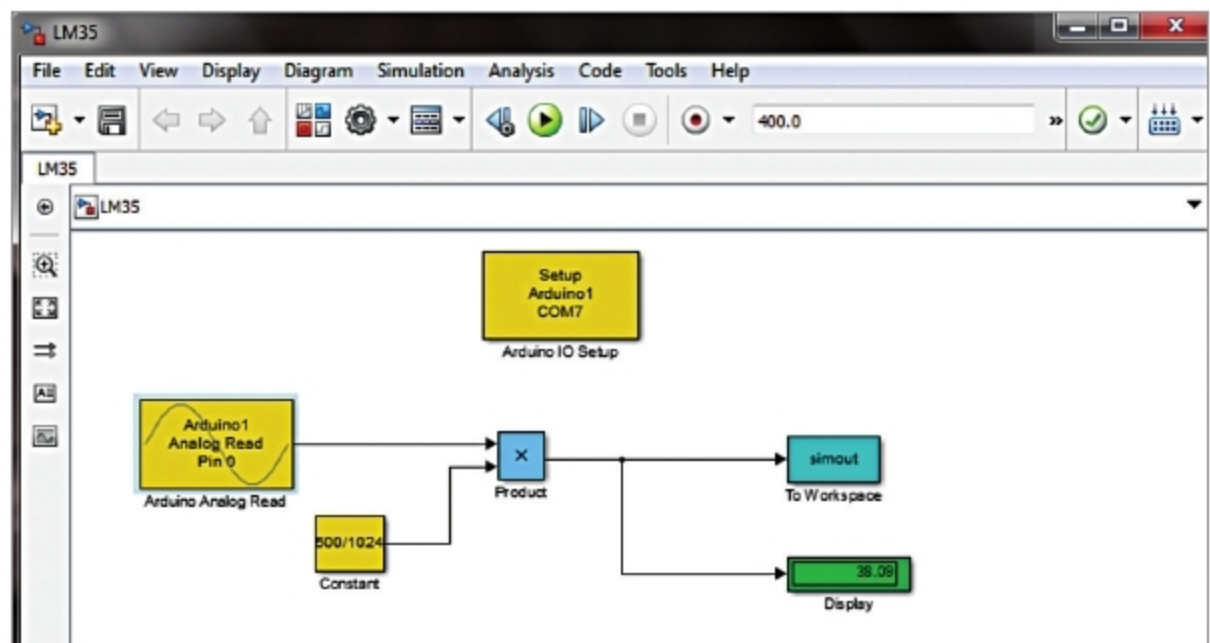


Figure 3: Screenshot of the Simulink model

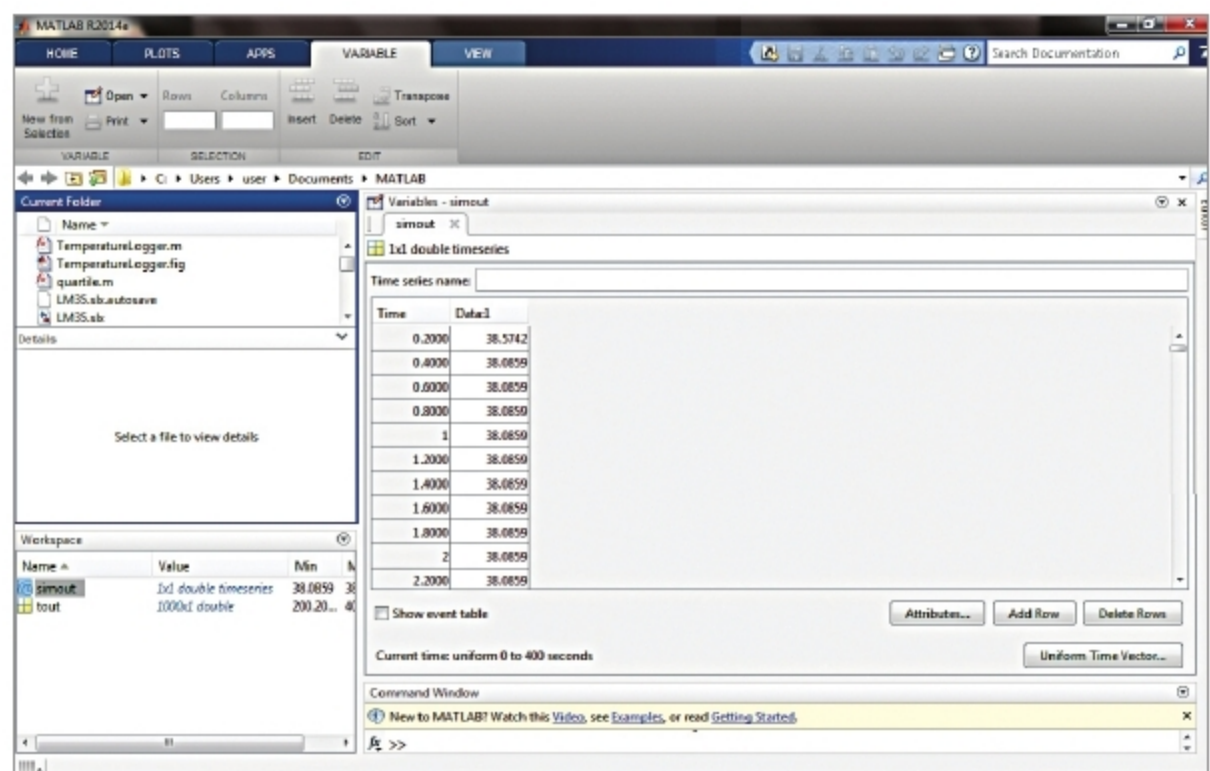


Figure 4: Screenshot of the temperature data stored in MATLAB workspace

number in your PC where the Arduino has been installed

- If you want to connect LM35 to some other analogue pin of Arduino, double click on 'Arduino Analogue Read' block and set the corresponding pin
- Set 'Simulation Stop Time' as per your requirement
- Click on 'Run' button in Simulink

The corresponding value of temperature can be seen on the 'Display' block in the model window. Additionally, the temperature data is also logged in MATLAB

workspace variable 'simout'. When the simulation terminates, the temperature data can be read by double-clicking on 'simout' in the workspace window of MATLAB. The screenshot of the temperature (in degC) data saved in MATLAB workspace is shown in Figure 4. **END** 🐧

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